Age- and Arm-Stratified Co-expression Network Rewiring in the Murine Kidney During and After Spaceflight

Ibrahim M. Shahid

Middle College, The University of North Carolina at Greensboro, USA
Corresponding author: imshahid@uncg.edu

Abstract. Astronauts exhibit elevated nephrolithiasis risk following spaceflight, and prior work links microgravity to distal nephron remodeling and renal transporter dysregulation. Whether spaceflight alters the regulatory coordination among genes — shifting co-expression architecture without necessarily changing mean expression levels — and whether such rewiring differs by age or by spaceflight habitat, remains uncharacterized. We addressed this using OSD-771 (RRRM-2), bulk kidney RNA-seq from 80 female C57BL/6NTac mice under a full $2 \times 4 \times 2$ factorial design crossing Age (young, 16 weeks; old, 34 weeks), Environment Group (FLT, GC, VIV, BSL), and Arm (ISS Terminal sacrifice, ISS-T; Live Animal Return, LAR). Nephron segment proportions were estimated via MuSiC deconvolution against a murine single-cell kidney reference and used to remove compositional confounding while preserving biological contrasts. Sample-specific co-expression networks were inferred via LIONESS on a shared sparse skeleton constructed with cell-standardized Ledoit-Wolf partial correlations, and edge-wise regression over the full factorial design yielded model-predicted contrast-specific networks. Node2vec embeddings of these networks, aligned by orthogonal Procrustes across multi-seed runs, produced per-gene rewiring scores (cosine distance) for four primary flight contrasts. Focused permutation testing identified 11 significantly rewired genes in young ISS-T animals (FDR < 0.05), including the renal amino acid transporter Slc6a20a, the circadian regulator Cry2, and the vitamin D-binding protein Gc, with two additional hits in old ISS-T animals (Prlr, Nr4a1). Pathway enrichment revealed coordinated ECM and complement remodeling in young ISS-T mice, an EMT and steroid biosynthesis signature in old ISS-T mice, and retinoic acid metabolism as the most statistically robust signal across both arms — with strongest enrichment in the LAR recovery condition (OR = 19.9, $q = 1.6 \times 10^{-5}$), suggesting active renal repair signaling during re-adaptation to terrestrial gravity. The ISS-T arm consistently showed greater and more significant rewiring than LAR across all contrasts, consistent with partial network recovery following return to Earth. Pre-registered NCC-WNK axis genes did not emerge among top individual rewiring hits, though pathway-level retinol and circadian signals offer candidate upstream mechanisms for known DCT dysregulation. This framework identifies retinoic acid signaling, ECM remodeling, and circadian network disruption as age-stratified transcriptomic responses to spaceflight invisible to standard differential expression analysis, and establishes a statistically defensible methodology for network rewiring inference in small-n factorial space biology datasets.

Key words: spaceflight; kidney; transcriptomics; co-expression networks; network rewiring; aging

1. Introduction

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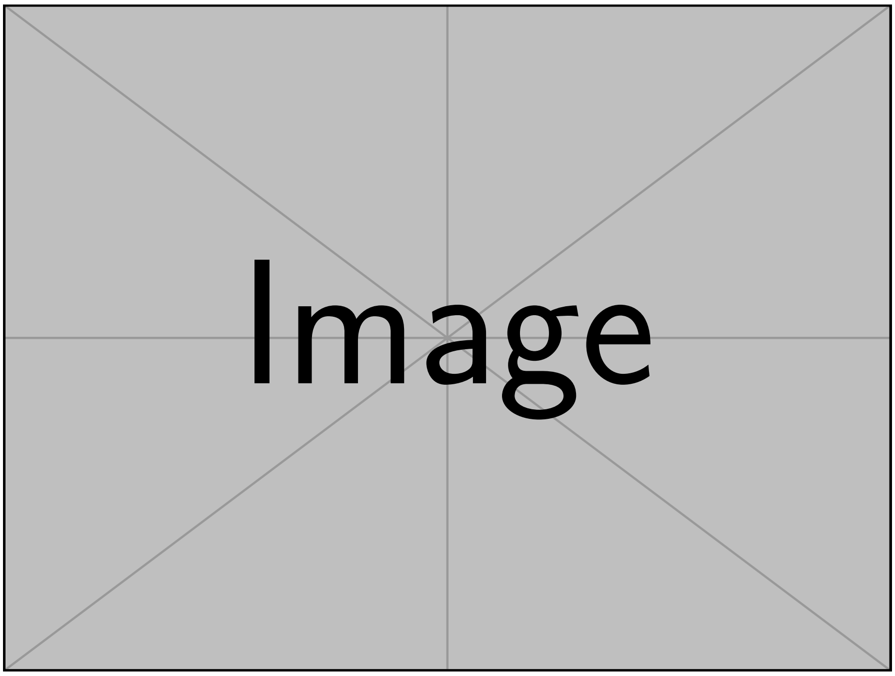


Figure 1: A sample figure (replace `example-image` with your own file).

Table 1: A sample table for demonstration.

Method	**Time (ms)**	**Accuracy (%)**
Baseline	12.5	91.2
Proposed	10.1	93.8
Optimized	8.7	94.4

We will refer to a classic text using a sample citation [1]. We also demonstrate inline math, e.g., $e^{i\pi} + 1 = 0$, and a displayed equation:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}. \tag{1}$$

1.1. A sample theorem

Theorem 1 (Toy result)**.** *For every integer $n \geq 1$, the sum $\sum_{k=1}^{n} k$ is equal to $\frac{n(n+1)}{2}$.*

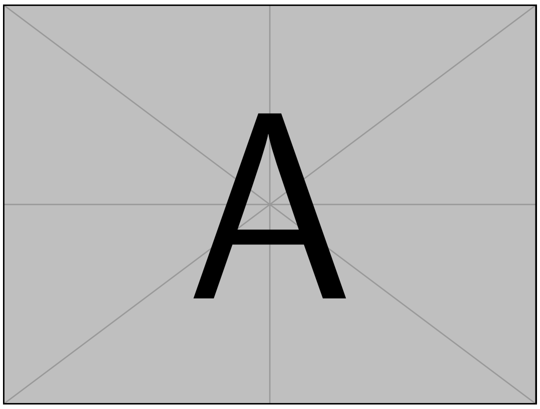
Proof. The proof can be done by induction. The base case $n = 1$ holds. Assume true for n. Then $\sum_{k=1}^{n+1} k = (\sum_{k=1}^{n} k) + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. □

2. Sample Figure(s)

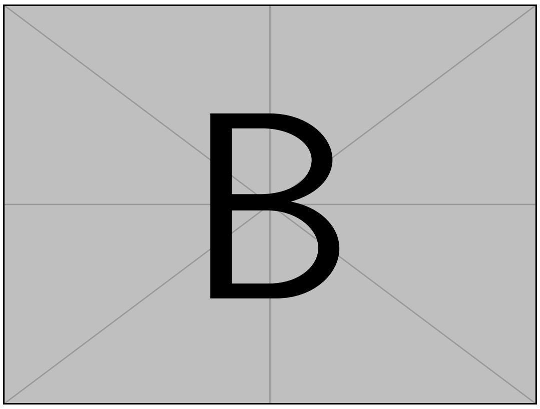
Figure 1 shows a sample figure. Figure 2 shows two subfigures.

3. Sample Table

Table 1 shows a simple table with aligned columns.



(a) First subfigure



(b) Second subfigure

Figure 2: A sample subfigure layout.

Algorithm 1 Sample procedure (toy example).

Require: Input array $A[1..n]$
Ensure: `true` if A is non-decreasing, else `false`
for $i \leftarrow 1$ to $n-1$ **do**
    if $A[i] > A[i+1]$ **then**
        return `false`
    end if
end for
return `true`

4. Sample Algorithm

Algorithm 1 shows a basic algorithm environment.

5. Conclusion

We included samples of an equation (Eq. 1), figures (Fig. 1–2), a table (Table 1), and an algorithm (Algorithm 1) to serve as a template.

References

[1] Donald E. Knuth. *The TeXbook*. Addison-Wesley, 1984.